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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of NITTE (Deemed to be University)
Second Semester M.Tech (EVT) (CBCS) Degree Examinations

May-June 2025

22EVT201 – POWER ELECTRONIC CONVERTERS

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Analyze the switching characteristics of a Power MOSFET.	8	L*3	1	1,2
	b) With the help of suitable example explain how the power conversion is made with the power switches.	7	L2	1	1,2
	c) Explain the importance of a filter in power supply circuit designs.	5	L2	1	1,2
2.	a) With the help of relevant circuit explain the concept of voltage-mode operation in controlling the output of the power supply.	10	L2	1	1,2
	b) Analyze the operation of Buck converter circuit for the CCM mode with relevant waveforms.	10	L3	1	1,2
Unit – II					
3.	a) Obtain the analysis for 12V, 4A Voltage mode Buck from a 28V source. The detailed specifications are below: The specification states that the output must be 12V when loaded by a 4A current and powered by a source varying from 20V to 30V, the maximum peak to peak output voltage ripple must always stay below 125mV and the switching frequency is 100kHz. Please note an AC input ripple current specification of 15mA peak to peak.	10	L3	2	1,2
	b) Explain the current Mode operation of Buck Converter operating in CCM. Also write the equations for Buck current mode equations.	10	L2	2	1,2
4.	a) Explain the component constraints for the Buck- Boost Converter for both CCM and DCM operation.	10	L2	2	1,2
	b) Obtain the design analysis for 948V, 2A Voltage Mode Boost from a car battery. A Boost converter is designed to deliver around 100W from a 12V car battery. The detailed specifications are below: $V_{in\ min} = 10V$, $V_{in\ max} = 15V$, $V_{out} = 48V$. $V_{ripple} = \Delta V = 250mV$, $V_{out\ drop} = 200mV$. Maximum from $I_{out} = 1$ to 10A in 10 μs . $I_{out\ max} = 2A$ $F_{sw} = 300kHz$.	10	L3	2	1,2
Unit – III					
5.	a) Write a note on (i) Hold Up time and (ii) In-Rush Current.	10	L3	3	1,2
	b) Show that RMS current in the capacitor is $I_{C\ Bulk,RMS,total} = \sqrt{I_{C\ bulk,RMS}^2 + I_{smpls,ac}^2}$	10	L4	3	1,2
6.	a) With neat circuit diagram explain the operation of Graetz bridge and obtain the expression for (i) input Voltage (ii) Depletion Current and (iii) Peak Recharge Capacitor Voltage.	10	L4	3	1,2
	b) Explain how active power factor correction can be obtained using Boost converter.	10	L3	3	1,2

P.T.O



Unit – IV

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|-------|---|----|----|---|-----|
| 7. a) | With neat sketch and associated waveforms obtain the analysis of an isolated Buck-Boost converter. | 10 | L4 | 4 | 1,2 |
| b) | With neat sketch and associated waveforms explain the two-switch configuration of a forward converter. | 10 | L3 | 4 | 1,2 |
| 8. a) | With neat SPICE Simulated circuit, analyze the Clamping of Drain Excursion of the network | 10 | L4 | 4 | 1,2 |
| b) | With neat sketch and associated waveforms explain the two-switch configuration of a Fly Back converter. | 10 | L3 | 4 | 1,2 |

Unit – V

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|--------|---|----|----|---|-----|
| 9. a) | Draw the neat sketches of any six types of bidirectional ac-dc converter topologies applicable in PHEV for battery charging. | 12 | L5 | 5 | 1,2 |
| b) | With neat circuit diagram explain the topology of bidirectional ac-dc converter as battery charger. | 8 | L3 | 5 | 1,2 |
| 10. a) | With neat operating circuit modes analyze the operation of a vehicle bidirectional dc-dc converter. Also obtain state space model for the same. | 12 | L5 | 5 | 1,2 |
| b) | Explain the four modes of EV charging as per IEC 61851. | 8 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
